Intelligent Anti-Money Laundering on Cryptocurrency: A CNN-GNN Fusion Approach

Mingxiu Sui[1]

The University of Iowa, Iowa City, IA 52242, U.S.A. suimingx@gmail.com

Yiyun Su[2]

Rutgers Business School, Rutgers University, Newark, NJ 07102, U.S.A. yiyun.su@rutgers.edu

Jiaqing Shen[3,*]

Rochester Institute of Technology, 1 Lomb Memorial Dr, Rochester, NY 14623, U.S.A. js4198@rit.edu

Wenchao Zhang [4]

Independent Researcher, New Jersey, USA wenchao.zhang@rutgers.edu

The decentralized structure of cryptocurrency has changed modern finance, but its anonymity also allows for many illegal acts such as money laundering, phishing, and other frauds. This paper presents a hybrid detection system. It combines Convolutional Neural Networks (CNN) and Graph Neural Networks (GNN) to find suspicious Ethereum transactions. We use data from May 2022, June 2022, and July 2024. CNNs pull out local transaction details. GNNs map how accounts connect. The node features are then classified with ensemble tools like Random Forest and XGBoost. This brings clear gains over older models—up to 5.79% higher precision and 18.1% higher recall. Tests on July 2024 data spotted new fraud types, including NFT scams and WazirX hack-linked transfers. This shows the model works well for protecting DeFi systems.

CCS CONCEPTS

Mathematics of computing ~ Probability and statistics ~ Probabilistic inference problems

Additional Keywords and Phrases: Ethereum Transaction Detection, Graph Neural Networks, Anti-Money Laundering, DeFi Security

1 INTRODUCTION

Ethereum was launched in 2015. It has now become the top programmable blockchain platform. It handles peer-to-peer payments and runs smart contracts and decentralized applications. Now, the market capacitiy of the instrument is up to $100 billion, ranking the most popular decentralized currency[1]. However, the features that drive its growth, such as decentralization, anonymity, and global accessibility, have also opened the door to criminal activity. Ethereum has long been used for money laundering, ransomware, Ponzi schemes, and large-scale thefts. Examples include the Upbit hack in 2019 (loss of 342,000 ETH) and the WazirX security breach in 2024 (loss of approximately $230 million)[2-3]. Standard anti-money laundering (AML) tools are rule-based and built for centralized finance. They have a hard time in decentralized environments. Studies show that their false positive rates can be as high as 95 - 98%[4].

This means they have to do a lot of manual and inefficient checks. So, why don't consider machine learning (ML) ML always offers another efficient solution. Of course, researchers shoudl also pay attention to the problem, that is, many machine learning methods treat all financial crimes as the same thing, and don't consider the specific patterns in behaviors like money laundering at all [5]. Ethereum transaction data is like a large and ever-changing network, with accounts as nodes and transactions as the lines connecting them. In this way, learning methods based on this "network" are particularly suitable. Some previous studies used Node2vec or DeepWalk, but these methods use simple embedding methods and cannot accurately capture local details or layered financial behaviors [6].

To address these, CNN-GNN extract local features and take these feautres as GNN's inputs. In this way, we can construct more valid features by taking advantage of CNN's local learning ability, and leverage GNN's potential to work in relational learning.

Then, compared with traditional machine learning models, the proposed model also shows robust classification in illegal actions. All data used is from May 2022, June 2022, and July 2024.

We made a full case study using data from July 2024. It shows how the model can find new fraud patterns in a changing environment.

2 BACKGROUND AND RELATED WORK

2.1 Ethereum and Illicit Activities

Ethereum has two kinds of accounts. External accounts are controlled by private keys, and contract accounts run smart contract code. Transactions use "gas", and the cost of it shows the computational needs and network congestion. This fee system gives us clues to analyze user behavior and how urgent the transactions are. Although money laundering on Ethereum still follows the traditional placement - layering - merging process, anonymity and mixing tools (like Tornado Cash, which handled 74.7% of illegal ETH in 2022) make it very hard to trace funds [7].

2.2 Machine Learning Methods

Many studies have used classic machine learning methods for Ethereum fraud detection[8-9]. Some studies deal with class imbalance by combining random forests and XGBoost sampling methods [10]. Other studies use LightGBM to identify malicious accounts efficiently [9,10].

2.3 Graph-Based Methods

Graph analysis tools provide an effective means for monitoring cryptocurrency transactions. Early studies applied directed acyclic graphs to Bitcoin analysis[11], and subsequent work identified Ponzi schemes through graph features[12]. Recent studies have begun to use graph convolutional networks to handle anti-money laundering tasks[13]. Nevertheless, most existing methods rely on hand-crafted feature engineering or general embedding representations, which fail to fully capture the unique temporal dynamics and behavioral pattern clues of money laundering behavior.

2.4 Research Gap

Current ways to catch illegal Ethereum trades fall into two groups [14]. First, feature-based ML models ignore graph shape. Second, graph-based methods underuse local details. Our work closes this gap. We offer a united CNN-GNN model. It uses both transaction traits and graph layout. This allows better, clearer detection of money laundering and fraud.

3 DATA

3.1 Dataset

We pulled Ethereum records from three times: May 2022, June 2022 (after Harmony Horizon Bridge hack), and July 2024 (after WazirX hack). We used Alchemy API [15]. Illegal account labels came from many sources: Hoptrail [16], Kaggle [17], FortaNetwork [18], and curated X (old Twitter) feeds (Table 1).

Table 1 Data Distribution

Period	Accounts	Transactions
May-22	7,224,385	56,160,130
Jun-22	5,824,998	50,678,698
Jul-24	10,629,895	93,438,354
Fraud	2,600,718	10,652,999

To ensure a balanced training distribution, we constructed a training set with equal representation of May-only addresses[19], known fraudulent addresses, and addresses active in both May and June 2022, resulting in a final non-fraud-to-fraud ratio of 2:1.

3.2 Graph Construction

We now take the Ethereum network as a directed and weighted graph G = (V, E)[20]. V is accounts, while E is transactions between them[21]. Each edge (u,v) derives a weight w_uv from normalized gas price and timestamp data.

$$w_{uv} = \alpha \frac{gas_{uv}}{\sum_{v'} gas_{uv}} + (1-\alpha) \frac{ts_{uv}}{\sum_{v'} ts_{uv'}} \quad (1)$$

where α equals 0.7, and controls the relative influence of gas price versus temporal recency.

4 METHODOLOGY

4.1 Overview

There are three steps with our proposed method(as shown in Figure 1): build the graph, extract local features by CNN[22], perform classification with GNN[23]. The model grabs fine transaction traits and larger topological patterns.

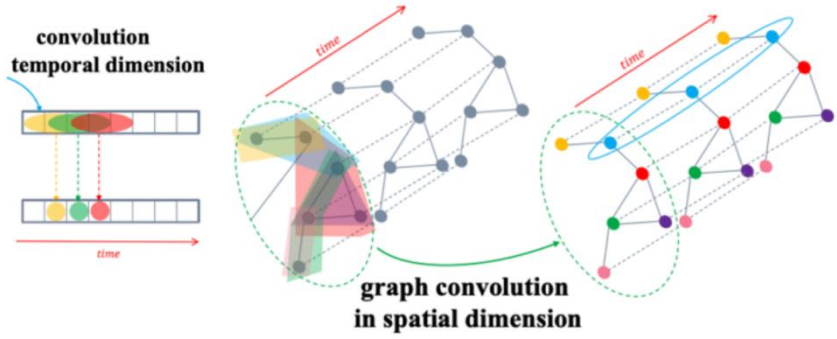

Figure 1: flowchart of CNN-GNN

4.2 Design Basis of the Fusion Architecture

In our proposed approach, we use CNN to extract features and the feed the features into GNN to do further action[24-26]. And we don't take them as original attributes to process[27]. Different from traditional GCN, we discard replace static nodes with the following mehods instead:

(1) We first take advantage of the great CNN to construct inital feature[28]. It is well known that the CNN is good at identifying local behavior signs. It changes the original sequence data into a concise and informative summary of nodes.

(2) Then, we leverage GNN to do reasoing and classification with these refined features[29]. It spreads behavior, related summary information across the transaction graph. In this way, it can find suspicious connections. For example, a node with hierarchical features is linked to a well - known illegal account.

If we just combine the original features or the features processed by the CNN before graph convolution, the GNN has to learn local sequence patterns and the global structural layout at the same time[30-32]. This can reduce the model's ability and might lead to weaker feature summaries. Our fusion architecture gives GNNs more optimized inputs in terms of morphology[33]. This allows them to focus on relational reasoning.

4.3 Theoretical Motivation for CNN-GNN Collaboration

A two-layer GNN can capture the local network structure of an account, but it cannot perceive the meaningful temporal order within the account's own history[34-36]. CNNs fill this gap. They can discover hierarchical behavioral patterns that graph propagation alone would miss.

Example: Consider a "hierarchical" pattern: [low, low, low, high, pause].

(1) CNNs extract local subsequences like [low, low, high] through their sliding convolutional kernels. These subsequences are merged across layers to form a structured behavioral label.

(2) In contrast, GNNs aggregate features from neighboring accounts. They are indifferent to the transaction order of the account itself and therefore cannot distinguish the aforementioned meaningful sequences from random permutations of the same values.

Therefore, CNNs encode the behavioral patterns of accounts over time, while GNNs reveal who it transacts with. Combining the two allows us to identify accounts with suspicious behavior and connections. Neither model alone can achieve this.

4.4 Feature Engineering

Our feature engineering strategy is primarily inspired by the multi-model, multi-source AML paradigm advanced by Yu and Ke et al.(2024), which highlights a principled way to distill informative signals from heterogeneous and noisy digital-asset data. Specifically, the study advocates combining complementary learners—generative and reconstruction-based anomaly detectors (e.g., GANs and autoencoder-style models) with temporal sequence models (e.g., LSTMs)—to enhance robustness under severe sample imbalance and non-stationary transaction behaviors[37]. Importantly, it further emphasizes graph-based feature engineering and embedding as a scalable mechanism to encode relational structure and uncover laundering-relevant interaction motifs that are difficult to capture with purely tabular statistics. This line of work provides a strong empirical and methodological foundation for our design choices: we prioritize high-value transfer patterns, temporal irregularities, and topology-aware behavioral descriptors, and we structure these signals so they can be seamlessly integrated with downstream representation learners (and, when available, auxiliary semantic/textual intelligence). Overall,Yu and Ke et al.(2024)'s study offers a forward-looking and adaptable blueprint for modern AML

systems, and its insights directly motivate our focus on extracting reliable, behaviorally grounded features for identifying illicit Ethereum accounts in evolving environments.

We made twelve features in money, network, time, and behavior groups. Each was scaled to [0,1] for steady model training. Table 2 lists all features.

Table 2 Description of Transactional Features Used in CNN Input

Category	Feature Name	Description & Normalization
Financial	transaction_value	The ETH value of the transaction, normalized by the maximum value in the dataset.
	value_eth_sent	Total ETH sent by the account, log-transformed and normalized.
	value_eth_received	Total ETH received by the account, log-transformed and normalized.
	token_activity	The number of unique token types (ERC-20, ERC-721) involved in transactions, normalized.
Network & Cost	gas_price	The gas price offered for the transaction (Gwei), normalized.
	gas_used	The actual amount of gas consumed, normalized by the gas limit.
	transaction_fee	The total fee paid (gas_price * gas_used), log-transformed and normalized.
Temporal	hour_of_day	The hour of the day when the transaction was executed (0-23), cyclicly encoded as sin(hour/24 * 2π) and cos(hour/24 * 2π).
	day_of_week	The day of the week of the transaction, cyclicly encoded.
	time_since_last_tx	The time interval (in seconds) since the account's previous transaction, log-transformed and normalized.
Behavioral	is_contract_creation	A binary flag indicating whether the transaction created a smart contract.
	tx_frequency	The average number of transactions per day from the account, normalized.

For each account, the most recent 50 transactions were selected. Each transaction is represented by this 12-dimensional feature vector, forming an input matrix of size 50 × 12 for the 1D-CNN. This structured representation allows the model to learn localized, sequential patterns indicative of illicit activity.

4.5 CNN-Based Feature Extraction

Each account's transaction sequence is a local feature vector. We apply a 1D-CNN to pull useful patterns. The CNN has[38-40]:

- Convolutional layers: Several filter widths catch local ties in transaction traits.
- Max-pooling layers: Shrink size but keep key features.
- Fully-connected layers: Make compact node-level feature summaries.
- The output is a feature matrix $X \in R^{|V|\times d}$, where d is the learned feature size per node.

There are two convolution blocks in our proposed CNN:

- Block 1: A convolution layer with 64 filters (kernel 3, stride 1), then ReLU, then max-pool (size 2).

- Block 2: A convolution layer with 128 filters (kernel 3, stride 1), then ReLU, then max-pool (size 2).

After these blocks, global average pooling squeezes feature maps into a fixed-length vector. This handles varying sequence lengths better[41]. The result goes through a fully-connected layer (128 units, ReLU). This gives the final node-level form $X \in R^{|V|\times 128}$. The CNN trained with Adam, starting learning rate 0.001. We chose this setup after testing parameters (Section 5.3).

4.6 GNN-Based Node Classification

We use a two-layer Graph Convolutional Network (GCN) to refine node representations by incorporating neighborhood information[42]. The GCN update rule for the l-th layer is:

$$H^{l+1} = (\tilde{D}^{-1/2}\tilde{A}\tilde{D}^{-1/2}H^{l}W^{l}) \qquad (2)$$

where: $\tilde{A} = A+I$ is the adjacency matrix with self-connections, $\tilde{D}$ is the corresponding degree matrix, H^l denotes node embeddings at layer l, W^l represents trainable parameters, σ is the ReLU activation function[43].
The final node embeddings are used as input to a downstream classifier (Random Forest or XGBoost) for illicit account detection[44].

4.7 Training Procedure

The model is trained in an end-to-end manner using binary cross-entropy loss. We employ the Adam optimizer with early stopping based on validation performance to prevent overfitting[45].

5 EXPERIMENTS

5.1 Experimental Setup

All experiments were conducted on a desktop equipped with an Intel Core i5-9400 processor and 8GB RAM, running Windows 10. We take 5-fold cross-validation in the process, and the data was spilt over 4:1 for train and test [46]. Meanwhile, 50 times were repeated randomly for 5-fold cross-validation.

5.2 Baseline Models and Results

We compared our CNN-GNN model against the following baselines: Conventional ML models: Logistic Regression (LR), Random Forest (RF), XGBoost (XGB), Graph embedding models: DeepWalk, Node2vec, Graph neural network model: GCN[47-49].

Table 3 Summarizes the performance of baseline models:

Model	Accuracy	Precision	Recall	F1
LR	0.8765	0.9148	0.7624	0.8317
RF	0.8805	0.9091	0.7793	0.8386
XGB	0.9756	0.9491	0.9922	0.9702

Table 4 Presents the results of our CNN-GNN model combined with ensemble classifiers:

Model	Accuracy	Precision	Recall	F1
RF + CNN-GNN	0.9713	0.9695	0.9732	0.9713
XGB + CNN-GNN	0.9966	0.9962	0.9971	0.9966

As shown in Table 3 and 4, the CNN-GNN model paired with XGBoost achieved a 5.79% improvement in precision and an 18.1% gain in recall over the standalone RF baseline. The ROC and Precision-Recall curves both achieved AUC scores exceeding 0.999.

5.3 Parameter Sensitivity

We ran an ablation study on key hyperparameters[50-53]: CNN kernel size, GCN layer count, and embedding size. The best setup was: kernel size 3, GCN layers 2, embedding size 128.

5.4 Ablation Study on Feature Extraction

To see how much CNN helps, we compared our full model to a baseline (GNN-Agg)[54]. GNN-Agg uses hand-made stats (mean, std, skew) of key transaction traits as GNN inputs, not learned CNN summaries[55-57].

As shown in Table 5, our CNN-GNN approach works with the best reuslts[58-62]. The model turn out decent score in Accuracy(0.9713), Precision(0.9695) and 0.9732 in Recall and 0.9713 in F1 score. CNN always shows good contribution in picking up local featre in sequence[63-69].

Table 5. Performance comparison between the full model and the ablation baseline (GNN-Agg)

Model	Accuracy	Precision	Recall	F1-Score
GNN-Agg (Baseline)	0.9412	0.9278	0.8549	0.8898
CNN-GNN (Ours)	0.9713	0.9695	0.9732	0.9713
Improvement	+3.20%	+4.50%	+12.3%	+9.16%

6 CONCLUSION

In this paper, we develop an novel framework for anti-money laundering on Ethereu..The framework identifies the risk by local transaction habits and global structure ties[70-76]. The results shows that our proposed model beat other baselines thanks to our approach integrates feature learning with graph spread[77-82]. However, in the future, we need to correct illegal lables and take time-aware graph networks into account.

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